

2 The following features of viruses can be used to classify them:

- the type of nucleic acid they contain
- the arrangement of the proteins in their capsid
- the presence or absence of an envelope
- the enzymes they contain.

(a) For each structure, put one cross ☒ in the appropriate box, in each row, to show which viruses have that structure.

(3)

Structure	Virus			
	Ebola only	HIV only	both Ebola and HIV	neither Ebola nor HIV
DNA	☐	☐	☐	☐
helical capsid	☐	☐	☐	☐
envelope	☐	☐	☐	☐

(b) Human cells contain several types of DNA polymerase.

One type of DNA polymerase, found in some cancer cells, has been shown to synthesise DNA from an RNA template.

(i) Name the enzyme, found in some types of virus, that can synthesise DNA from an RNA template.

(1)

(ii) Suggest why this DNA polymerase may be a target for drugs used to treat cancer.

(1)

(Total for Question 2 = 5 marks)